just as (or sometimes more!) important. Elixir and the BEAM provide a core foundation for programming the unhappy path you want, rather than taking whatever you get.

Technical Benefits

Elixir applications can fully utilize their host CPU cores (parallelism) without having to write complex code. It's possible for other languages to achieve this same level of parallelism, but the unique aspect is how simple it is to achieve this with Elixir—it's nearly free.

It's easy to overlook the impact of this. If your web application is served on a 4-core server and you increase it to an 8-core server, you'll likely get twice the throughput out of it without making any changes. Despite it seeming like it should be common in other languages, it's not. Many languages (including Ruby) become bound as the server size grows, and applications can't utilize all of their cores when that happens.

Parallelism is crucial for modern programming, but the opposite of parallelism is just as important—serial code. Elixir provides excellent ways to control when code executes in parallel and when it executes serially.

The combination of parallel and serial code gives you complete control over your system runtime performance. This is unmatched in almost any other programming language today. Elixir (and BEAM languages) are equipped with these world-class capabilities out of the box.

Elixir is well-suited for building reliable, modern systems. This especially applies to web systems, but Elixir can be used in hardware systems, communication systems, machine learning, and more. You'll write more performant systems that are cheaper to run when you use Elixir.

These are only a few of the technical benefits of Elixir. A lot of Elixir's greatest strengths come from its foundation with Erlang and the BEAM. Let's dive into what those are next.

Erlang, OTP, Elixir, and Friends

Elixir stands on the shoulders of giants that have been around for decades, like Erlang, OTP, and the BEAM. You can be confident that Elixir's runtime is world-class, even though it's new when compared to other languages.

You don't have to know a ton about Erlang, OTP, and the BEAM to use Elixir, but you'll run into the edges of them at some point. For example, when installing Elixir you also had to ensure that the proper Erlang version was installed.